Tara Dacunha

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Research Interests

Keywords: *Dark matter (353); Dwarf galaxies (416); Stellar streams (2166)*

- Constraining dark matter physics with near field astrophysical observations
- Modeling populations of dark matter subhalos, dwarf galaxies, and tidal debris
- Flexible modeling using cosmological simulations, analytic methods, galaxy-halo models, star-tagging

Education

Stanford University 2022 – 2027 (projected)
Ph.D. in Physics
Advisor: Risa Wechsler
GPA: 3.99/4.00
Awards/Honors: *KIPAC Graduate Fellowship* (2023), *2MB Graduate Fellowship* (2026)

University of Pennsylvania 2018 – 2022
B.A. with Honors in Physics (concentration in astrophysics), Minor in Linguistics
M.S. in Physics
Research Mentor: Bhuvnesh Jain
GPA: 4.00/4.00 (B.A.) 3.91/4.00 (M.S.)
Awards/Honors: William E. Stephens Memorial Prize, ΦBK, *summa cum laude*

Teaching & Mentoring

- Research mentor, Stanford undergraduate student Reid Kanoa-Hammer 2026
- TA & mentor, Stanford Program for Inspiring the Next Generation of Women in Physics 2025
- TA, Stanford PHYSICS 191: Communication in Physics (rated 4.64/5 in effectiveness) 2025
- Lab TA, Stanford PHYSICS 42: Classical Mechanics Lab (rated 4.82/5 in effectiveness) 2023
- TA, Stanford PHYSICS 43: Electricity & Magnetism (rated 4.96/5 in effectiveness) 2023
- TA, University of Pennsylvania ASTRO 007: The Big Bang and Beyond 2021
- Tutor, middle and high school level math, physics, biology, chemistry, and SAT prep 2017-2022

Outreach & Service

- KIPAC Communications Committee member & designer 2026
- Guest Lecturer, Stanford Program for Inspiring the Next Generation of Women in Physics 2026
- Interview panelist for KIPAC communications associate 2026
- Artist and science communication volunteer, KIPAC Research Highlights 2026
- Counselor, Stanford Program for Inspiring the Next Generation of Women in Physics 2025
- Volunteer, KIPAC High School Solar Physics Outreach 2025
- Booth leader & activity developer, KIPAC Community Day Dark Matter Booth 2025
- Volunteer, Equity & Inclusion Panel, Stanford Physics, Identity, & Equity program 2024
- Volunteer, KIPAC Community Day Cosmic Magnetism Booth 2023

Publications

Cited in ∼151 academic works. *h*-index of 6.

7. Vedant Chandra, Charlie Conroy, Risa Wechsler, Ana Bonaca, Andrew Benson, Jacob Nibauer, **Tara Dacunha**, Philip Mansfield, Paul Menker, and Anya Phillips. Which Milky Way Streams Make the Best Dark Matter Detectors? . *arXiv e-prints*, page arXiv:2610.00456, September 2026
6. **Tara Dacunha**, Phil Mansfield, and Risa H Wechsler. Memoirs of mass accretion: probing the edges of intracluster light in simulated galaxy clusters . *The Astrophysical Journal*, 994(2):274, 2025
5. **Tara Dacunha**, Sergio Martin-Alvarez, Susan E Clark, and Enrique Lopez-Rodriguez. The fallibility of equipartition magnetic field strengths from synchrotron emission using synthetically observed galaxies . *The Astrophysical Journal*, 980(2):197, 2025
4. Sergio Martin-Alvarez, Enrique Lopez-Rodriguez, **Tara Dacunha**, Susan E Clark, Alejandro S Borlaff, Rainer Beck, Francisco Rodríguez Montero, Seoyoung L Jung, Julien Devriendt, Adrianne Slyz, and others. Extragalactic magnetism with SOFIA (SALSA Legacy Program). VII. A tomographic view of far-infrared and radio polarimetric observations through MHD simulations of galaxies . *The Astrophysical Journal*, 966(1):43, 2024
3. **Tara Dacunha**, Marco Raveri, Minsu Park, Cyrille Doux, and Bhuvnesh Jain. What does a cosmological experiment really measure? Covariant posterior decomposition with normalizing flows . *Physical Review D*, 105(6):063529, 2022
2. **Tara Dacunha**, Matthew Belyakov, Susmita Adhikari, Tae-hyeon Shin, Samuel Goldstein, and Bhuvnesh Jain. Connecting galaxy evolution in clusters with their radial profiles and phase space distribution: results from the IllustrisTNG hydrodynamical simulations . *Monthly Notices of the Royal Astronomical Society*, 512(3):4378–4393, 2022
1. Tae-hyeon Shin, Bhuvnesh Jain, Susmita Adhikari, Eric J Baxter, C Chang, S Pandey, A Salcedo, DH Weinberg, A Amsellem, N Battaglia, and others. The mass and galaxy distribution around SZ-selected clusters . *Monthly Notices of the Royal Astronomical Society*, 507(4):5758–5779, 2021

In-Prep Publications

1. **Tara Dacunha**, Phil Mansfield, Risa H Wechsler, and Ethan Nadler. SubMERGE: Subhalo Mass Evolution in Realistic Galactic Environments (to be submitted to ApJ 2026)

Selected Presentations

- Theoretical Astrophysics Center (TAC) Seminar, UC Berkeley (Invited) 2026
- Probe Combination for Dark Matter Physics Workshop, KICP, University of Chicago 2026
- Via All-hands Virtual Meeting: SubMERGE Model Update 2026
- Near Field Cosmology in the Era of Big Data, The Fields Institute 2026
- European Astronomical Society Meeting, Special Session 1 2026
- Via All-hands Workshop Lightning Talk, Harvard & Smithsonian Center for Astrophysics 2026
- Via All-hands Virtual Meeting: Subhalo Mass Loss Modeling 2026
- GFC Group, KIPAC, Stanford University 2026
- Via All-hands Workshop Talk on Stellar Stream Outreach, Yale University 2025
- IAU Symposium 403: The Hidden Beauty of the Galactic Outskirts (Poster) 2025
- KIPAC Tea Talk, KIPAC, Stanford University 2025